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GRADE A+
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

**Title: Importing business datasets from flat files & MySQL and
connecting with any other sources into Power BI**

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in Excel files, customer data in CSV files, and product information in another source. Management wants a centralized reporting system. Import data from multiple sources into Power BI, establish the required connections, and prepare the data for business reporting.

Introduction

In today's business environment, organizations often store data across multiple platforms — spreadsheets, flat files, and relational databases — making it difficult to get a single, unified view of performance. Power BI addresses this challenge by allowing data from different sources to be imported, cleaned, modeled, and visualized within one platform. This experiment simulates a real-world retail scenario where sales, customer, and product data originate from different systems, and demonstrates the complete workflow of consolidating them into a single interactive business dashboard.

Methodology / Steps Followed

Step 1: Data Collection

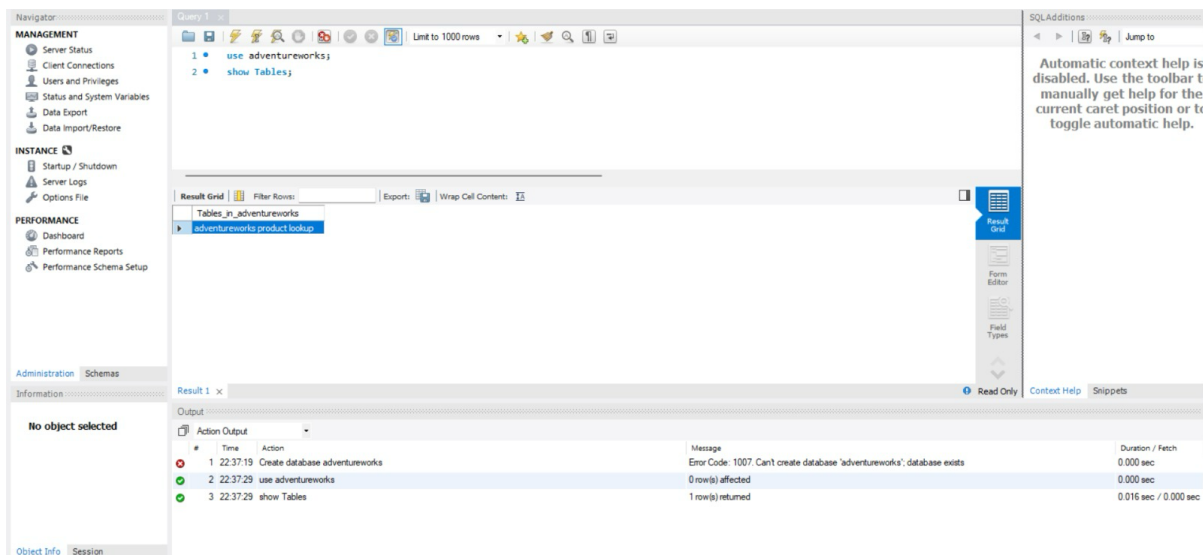
The data required for this business analysis was collected from multiple sources to build a centralized reporting model, reflecting how a real retail organization typically stores its information across departments and systems.

- Sales data was obtained from the AdventureWorks Sales Data 2022 Excel file, containing transactional-level records such as order dates, quantities, and product/customer references.
- Customer information was obtained from the AdventureWorks Customer Lookup dataset, containing demographic details such as name, gender, marital status, and income.
- Product information was obtained from the AdventureWorks Product Lookup dataset, sourced from a MySQL database rather than a flat file.
- Product category information was obtained from the AdventureWorks Product Categories Lookup dataset.
- Product subcategory information was obtained from the AdventureWorks Product Subcategories Lookup dataset.
- Regional/territory information was obtained from the AdventureWorks Territory Lookup dataset.

Collecting data in this fragmented way reflects how sales, customer, and inventory systems in real businesses are rarely unified, and highlights why a consolidation tool like Power BI is necessary for effective reporting.

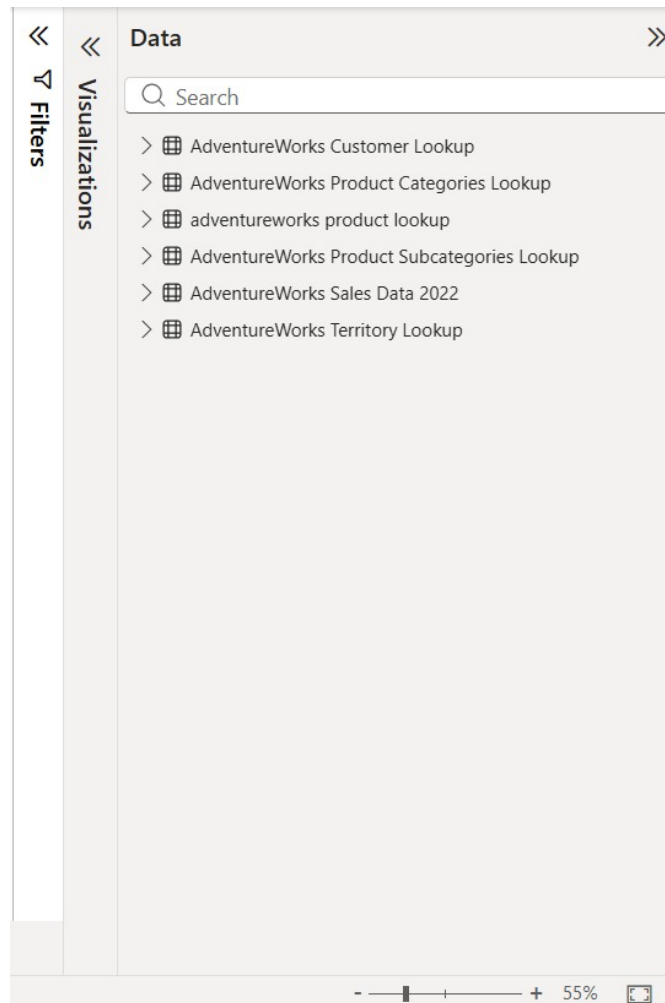
Step 2: Product Data Management Using MySQL

- Set up a MySQL database named "adventureworks" to manage the Product Lookup dataset independently of the other Excel-based sources.
- Verified the database and its tables using the queries `USE adventureworks;` and `SHOW TABLES;` in MySQL Workbench.
- Confirmed the Product Lookup table was available in the database, containing key fields such as ProductKey, ProductName, ProductSKU, ProductColor, ProductCost, ProductPrice, and ProductSize.
- Used this MySQL database as one of the data sources connected to the Power BI project.



Step 3: Data Import into Power BI

- Opened Power BI Desktop and created a new project file.
- Imported the Sales Data 2022, Customer Lookup, Product Categories Lookup, Product Subcategories Lookup, and Territory Lookup datasets using the Get Data option.
- Connected Power BI to the MySQL database using the Navigator window to import the Product Lookup table.
- Loaded all six tables into the Power BI data model.



Step 4: Data Cleaning and Transformation

All transformation was carried out in Power Query Editor before loading the data into the model, ensuring the underlying data was consistent and analysis-ready.

- Promoted headers and changed column data types across all tables — converting date fields to Date type, key columns to Whole Number, and descriptive fields to Text type.
- Cleaned the Customer Lookup table by filtering out irrelevant rows and removing blank or bottom rows to ensure only valid, complete customer records remained.
- Verified column-level data quality using Power Query's column profiling, confirming 0% error and empty values across the cleaned columns.
- Removed unnecessary columns that were not required for reporting.

Queries [6]

Table.TransformColumnTypes(#"Promoted Headers",{{"ProductCategoryKey", Int64.Type}, {"CategoryName", type text}})

ProductCategoryKey	CategoryName
1	Bikes
2	Components
3	Clothing
4	Accessories

2 COLUMNS, 4 ROWS Column profiling based on top 1000 rows

Queries [6]

Table.TransformColumnTypes(#"Promoted Headers",{{"CustomerKey", Int64.Type}, {"Prefix", type text}, {"FirstName", type text}, {"LastName", type text}, {"BirthDate", type date}, {"MaritalStatus", type text}, {"Gender", type text}})

CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus	Gender
1	11000	MR.	JON	YANG	08-04-1966	M
2	11001	MR.	EUGENE	HUANG	14-05-1965	S
3	11002	MR.	RUBEN	TORRES	12-08-1965	M
4	11003	MS.	CHRISTY	ZHU	15-02-1968	S
5	11004	MRS.	ELIZABETH	JOHNSON	08-08-1968	S
6	11005	MR.	JULIO	RUIZ	05-08-1965	S
7	11007	MR.	MARCO	MEHTA	09-05-1964	M
8	11008	MRS.	ROBIN	VERHOFF	07-07-1964	S
9	11009	MR.	SHANNON	CARLSON	01-04-1964	S
10	11010	MS.	JACQUELYN	SUAREZ	06-02-1964	S
11	11011	MR.	CURTIS	LU	04-11-1963	M
12	11012	MRS.	LAUREN	WALKER	18-01-1968	M
13	11013	MR.	IAN	JENKINS	06-08-1968	M
14	11014	MRS.	SYDNEY	BENNETT	09-05-1968	S
15	11015	MS.	CHLOE	YOUNG	27-02-1979	S
16	11016	MR.	WYATT	HILL	28-04-1979	M
17	11017	MRS.	SHANNON	WANG	26-06-1944	S
18	11018	MR.	CLARENCE	RAI	09-10-1944	S
19	11019	MR.	LUKE	LAL	07-03-1978	S
20	11020	MR.	JORDAN	KING	20-09-1978	S
21	11021	MS.	DESTINY	WILSON	03-09-1978	S
22	11022	MR.	ETHAN	ZHANG	12-10-1978	M
23	11023	MR.	SETH	EDWARDS	11-10-1978	M
24	11024	MR.	RUSSELL	XIE	17-09-1978	M
25	11025		ALEJANDRO	BECK	23-12-1945	M
26	11026	MR.	HAROLD	SAI	03-04-1946	S
27	11027	MR.	JESSIE	ZHAO	07-12-1946	M
28						

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

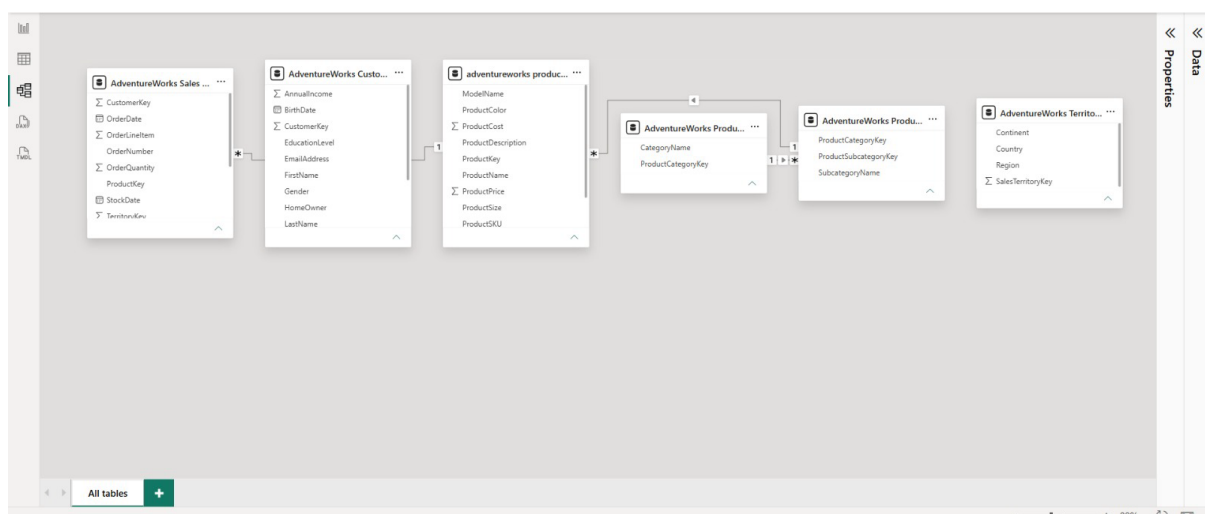
Queries [6] Table.TransformColumnTypes("#Promoted Headers",{"OrderDate", type date}, {"StockDate", type date}, {"OrderNumber", type text},

	OrderDate	StockDate	OrderNumber	ProductKey	CustomerKey	TerritoryKey	OrderLineItem
1	01-01-2022	13-12-2021	SO61285		529	23791	1
2	01-01-2022	24-09-2021	SO61285		214	23791	1
3	01-01-2022	04-09-2021	SO61285		540	23791	1
4	01-01-2022	28-09-2021	SO61301		529	16747	1
5	01-01-2022	21-10-2021	SO61301		377	16747	1
6	01-01-2022	23-10-2021	SO61301		540	16747	1
7	01-01-2022	04-09-2021	SO61269		215	11792	4
8	01-01-2022	21-10-2021	SO61269		229	11792	4
9	01-01-2022	24-10-2021	SO61286		528	11530	6
10	01-01-2022	27-09-2021	SO61286		536	11530	6
11	01-01-2022	23-10-2021	SO61298		530	18155	10
12	01-01-2022	02-12-2021	SO61298		214	18155	10
13	01-01-2022	15-12-2021	SO61298		223	18155	10
14	01-01-2022	01-10-2021	SO61310		538	13541	8
15	01-01-2022	08-11-2021	SO61310		584	13541	8
16	01-01-2022	06-12-2021	SO61270		485	18259	9
17	01-01-2022	31-10-2021	SO61289		528	23694	1
18	01-01-2022	16-11-2021	SO61289		536	23694	1
19	01-01-2022	20-11-2021	SO61289		215	23694	1
20	01-01-2022	12-10-2021	SO61278		477	22308	10
21	01-01-2022	28-09-2021	SO61278		479	22308	10
22	01-01-2022	10-12-2021	SO61278		488	22308	10
23	01-01-2022	25-09-2021	SO61278		580	22308	10
24	01-01-2022	01-10-2021	SO61305		583	20236	9
25	01-01-2022	13-09-2021	SO61283		529	26273	1
26	01-01-2022	27-09-2021	SO61283		538	26273	1
27	01-01-2022	03-09-2021	SO61283		220	26273	1
28							

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Step 5: Data Modeling

- Switched to Model view to connect all six tables into a single, unified data model.
- Connected the Sales Data 2022 table to the Customer Lookup table using CustomerKey.
- Connected the Sales Data 2022 table to the Product Lookup table using ProductKey.
- Connected the Sales Data 2022 table to the Territory Lookup table using TerritoryKey.
- Connected the Product Lookup table to the Product Subcategories Lookup table using ProductSubcategoryKey, and the Product Subcategories Lookup table to the Product Categories Lookup table using ProductCategoryKey, forming a snowflake-style hierarchy for product classification.
- Verified all relationships were one-to-many, with the Sales Data 2022 table acting as the central fact table.



Step 6: DAX Measures

Created DAX measures on the Sales Data 2022 table to calculate key business KPIs:

- Total Sales
- Total Orders
- Total Customers
- Total Quantity Sold
- Average Order Value

Total Sales = SUM('AdventureWorks Sales Data 2022'[SalesAmount])

Total Orders = DISTINCTCOUNT('AdventureWorks Sales Data 2022'[OrderNumber])

Total Customers = DISTINCTCOUNT('AdventureWorks Customer Lookup'[CustomerKey])

Total Quantity Sold = SUM('AdventureWorks Sales Data 2022'[OrderQuantity])

Average Order Value = DIVIDE([Total Sales], [Total Orders])

Step 7: Dashboard Development

- Developed a "Retail Sales Performance Dashboard" using Power BI, spread across two report pages.
- Created KPI cards for Total Sales, Total Orders, and Total Customers.
- Built a bar chart showing Total Sales by Region.
- Built a column chart showing Total Sales by CategoryName.
- Built a bar chart showing Total Sales by ProductName.
- Built a line chart showing Total Sales by Year, Quarter, Month, and Day.

Step 8: Interactive Filters and Slicers

- Added a slicer for OrderDate.
- Added a slicer for Region.
- Added a slicer for CategoryName.
- These slicers allow users to dynamically filter the entire dashboard by time period, geography, and product category.

Step 9: Dashboard Formatting

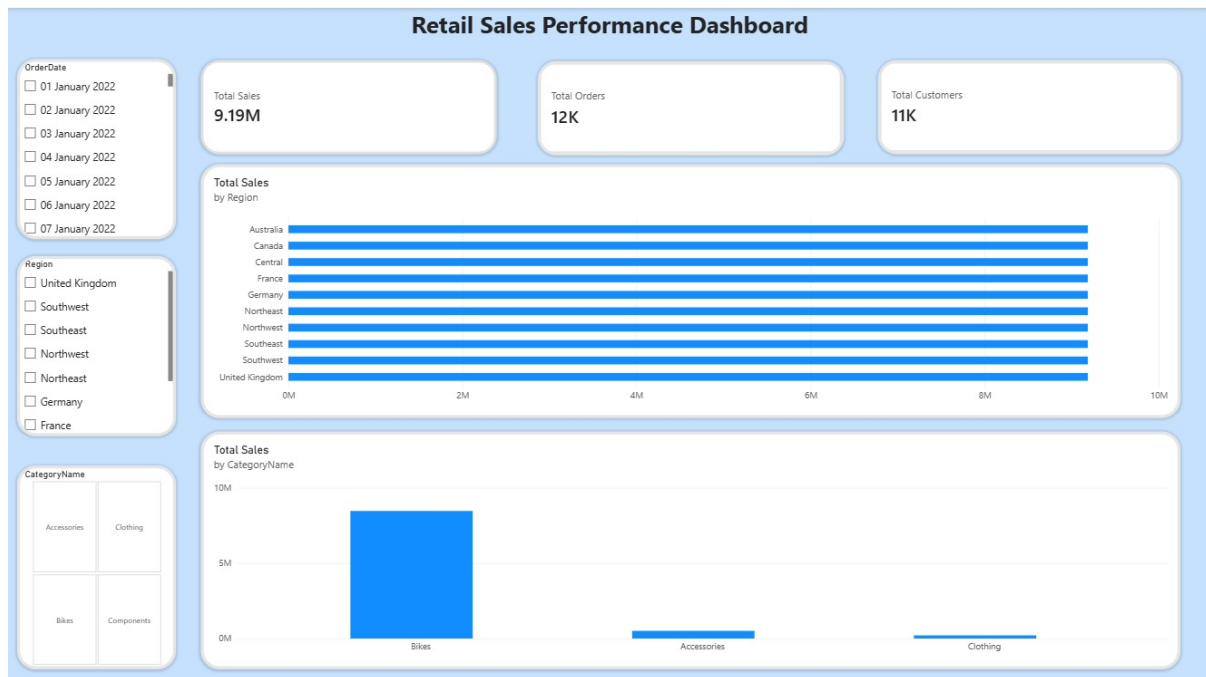
- Added the title "Retail Sales Performance Dashboard" at the top of the report.
- Arranged KPI cards, charts, and slicers in a structured, easy-to-scan layout.
- Applied a light background color to the canvas for a clean, professional appearance.
- Formatted sales values with appropriate number formatting (M for millions, K for thousands).
- Adjusted chart titles, axis labels, and visual sizing for clarity and consistency.

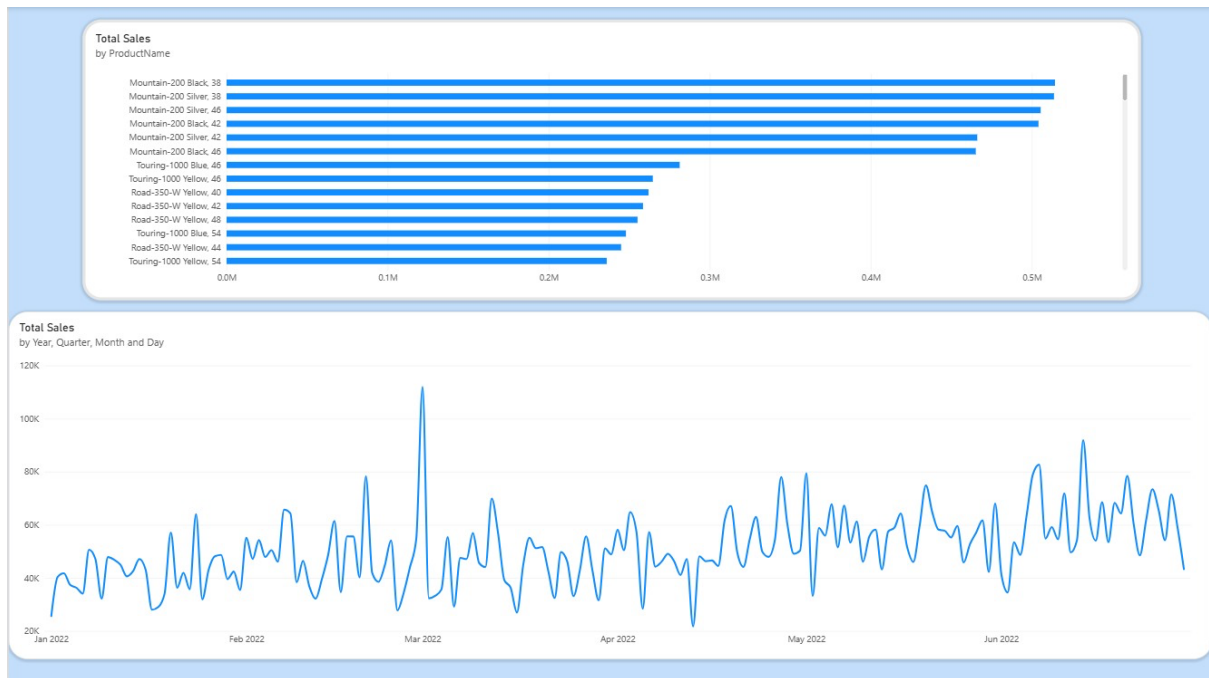
Step 10: Analysis and Insights

- Overall sales performance across the full year (2022), totaling approximately 9.19M in sales.
- Order and customer volume, with roughly 12K total orders placed by around 11K unique customers.
- Regional sales performance across countries such as the United Kingdom, Australia, Canada, Germany, and France.
- Category-wise contribution to sales, with Bikes emerging as the dominant category over Accessories and Clothing.
- Top-performing individual products, led by the Mountain-200 series.
- Daily and monthly sales trends across the year, including a notable peak in early March.

Step 11: Final Dashboard

After completing data integration, transformation, modeling, DAX calculations, and visualization, the final Retail Sales Performance Dashboard was developed across two pages.





Together, these pages provide a centralized and interactive view of business performance, enabling users to filter and analyze sales by date, region, category, and product — supporting faster, more informed business decisions.

Conclusion

This project successfully integrated data from six sources — Excel files, and a MySQL database for product data — into a single Power BI reporting model. The data was cleaned, transformed, and modeled into a snowflake schema connecting Sales, Customer, Product, Product Subcategory, Product Category, and Territory tables. DAX measures were developed to calculate key performance indicators such as Total Sales, Total Orders, Total Customers, and Average Order Value. The resulting Retail Sales Performance Dashboard presents this information through KPI cards, regional and category breakdowns, product-wise analysis, and monthly trend charts, supported by interactive slicers — providing a centralized view that supports data-driven business decisions.

Learning Outcomes

Through this project, I learned:

1. Data Collection and Integration — how to work with data from multiple Excel sources and a MySQL database.
2. MySQL Database Management — how to verify and manage a database and its tables using SQL commands.
3. Power BI Data Import — how to connect Excel and MySQL sources to Power BI and load multiple related tables.

4. Data Cleaning and Transformation — how to use Power Query to promote headers, fix data types, and remove invalid rows.
5. Data Modeling — how to build a snowflake-schema data model connecting fact and dimension tables through common keys.
6. DAX — how to create measures for KPIs such as Total Sales, Total Orders, and Average Order Value.
7. Data Visualization — how to build KPI cards, bar charts, column charts, and trend lines.
8. Interactive Dashboard Development — how to use slicers to let users filter by date, region, and category.
9. Business Analysis — how to interpret sales trends, top products, and regional performance from a dashboard.
10. Data-Driven Decision Making — practical understanding of how BI tools turn raw transactional data into actionable insights.